

KARLEBY-JAKOBSTAD, Finland

WMO: 029030

Lat: 63.718N

Lon: 23.139E

Elev: 25

StdP: 101.02

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-26.1	-22.1	-29.0	0.3	-26.1	-24.8	0.4	-22.4	10.5	-2.0	9.5	-1.6	0.8	30	0.667

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.3	26.9	18.2	24.9	17.2	22.9	16.2	19.6	24.4	18.4	22.8	17.4	21.5	4.0	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.0	13.0	21.3	16.9	12.1	20.5	15.5	11.1	19.1	55.9	24.7	52.1	22.6	48.9	21.5	24.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.1	8.0	7.2	DB	-29.0	29.1	4.6	2.5	-32.3	30.9	-35.0	32.4	-37.6	33.8	-41.0	35.7	(4)
(5)			WB	-29.8	20.7	4.9	1.7	-33.3	21.9	-36.1	22.9	-38.9	23.9	-42.4	25.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	3.8	-7.2	-7.3	-3.6	2.6	8.0	12.8	16.4	14.4	9.7	3.7	-0.7	-3.9	(6)
(7)		DBStd	9.42	7.06	7.12	5.33	3.72	4.30	3.57	3.45	3.29	3.56	4.02	4.80	6.13	(7)
(8)		HDD10.0	2762	532	484	420	224	90	12	1	4	48	196	321	431	(8)
(9)		HDD18.3	5328	791	717	678	473	320	169	81	128	258	453	571	689	(9)
(10)		CDD10.0	506	0	0	0	2	28	97	198	140	39	2	0	0	(10)
(11)		CDD18.3	30	0	0	0	0	1	4	19	6	0	0	0	0	(11)
(12)		CDH23.3	342	0	0	0	0	19	51	208	64	0	0	0	0	(12)
(13)		CDH26.7	63	0	0	0	0	1	11	42	9	0	0	0	0	(13)
(14)	Wind	WSAvg	3.3	3.3	3.3	3.4	3.3	3.5	3.4	2.9	2.9	3.1	3.2	3.4	3.4	(14)
(15)	Precipitation	PrecAvg	549	34	29	27	26	42	49	71	68	51	58	48	45	(15)
(16)		PrecMax	726	70	54	57	67	78	97	122	146	110	118	103	106	(16)
(17)		PrecMin	392	9	2	4	6	8	16	16	26	14	14	7	10	(17)
(18)		PrecStd	81	16	14	13	16	20	23	27	33	28	23	22	21	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.8	5.9	9.2	18.1	26.0	28.1	30.1	27.9	21.4	14.0	10.1	6.6	(19)
(20)			MCWB	3.7	3.5	4.6	10.0	15.2	18.3	20.0	18.2	15.9	11.6	8.9	6.0	(20)
(21)		2%	DB	3.1	3.8	6.8	14.4	22.8	24.2	27.8	25.1	19.0	12.0	8.2	5.2	(21)
(22)			MCWB	2.2	2.3	3.6	8.2	13.7	15.8	19.1	17.8	14.5	10.5	7.4	4.4	(22)
(23)		5%	DB	2.1	2.1	4.8	11.8	19.2	22.1	25.8	23.0	17.1	10.9	6.9	4.1	(23)
(24)			MCWB	1.4	1.2	2.3	6.8	11.9	14.7	18.0	16.8	13.5	9.6	6.3	3.4	(24)
(25)		10%	DB	1.1	1.1	3.1	9.3	16.2	20.2	23.8	21.0	15.8	9.8	5.4	2.9	(25)
(26)			MCWB	0.7	0.5	1.3	5.4	10.5	13.9	17.3	16.2	12.9	8.8	4.8	2.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.7	3.7	5.4	11.3	16.3	19.3	21.8	20.9	16.6	12.3	9.1	6.1	(27)
(28)			MCD B	4.1	5.2	8.3	15.9	23.9	25.0	27.4	24.9	19.9	13.5	9.7	6.2	(28)
(29)		2%	WB	2.3	2.3	3.7	9.0	14.5	17.4	20.2	18.9	15.3	11.0	7.5	4.6	(29)
(30)			MCD B	2.8	3.3	5.8	13.7	21.0	22.5	25.3	22.6	18.2	12.0	8.1	4.9	(30)
(31)		5%	WB	1.4	1.4	2.6	7.1	12.7	15.8	19.1	17.7	14.2	9.9	6.1	3.4	(31)
(32)			MCD B	1.8	2.0	4.2	10.8	18.7	20.5	24.0	21.3	16.4	10.6	6.5	3.9	(32)
(33)		10%	WB	0.6	0.5	1.5	5.6	11.0	14.7	18.0	16.7	13.3	8.7	4.9	2.2	(33)
(34)			MCD B	1.0	0.9	2.7	8.9	15.6	19.1	22.5	20.1	15.2	9.4	5.2	2.7	(34)
(35)	Mean Daily Temperature Range		MDBR	6.3	7.0	8.2	9.3	11.0	10.0	10.3	10.0	8.8	6.0	4.5	5.3	(35)
(36)		5% DB	MCD BR	4.7	5.5	8.8	13.8	16.6	13.9	14.0	13.2	10.9	6.2	4.8	4.7	(36)
(37)			MCWBR	4.4	4.6	6.2	8.5	8.8	7.0	6.6	6.9	4.9	4.5	4.2	(37)	
(38)		5% WB	MCD BR	4.8	4.9	7.4	12.3	14.9	12.0	12.1	11.0	9.5	5.7	4.6	4.6	(38)
(39)			MCWBR	4.7	4.3	5.6	8.0	8.2	6.7	6.2	6.4	4.7	4.5	4.3	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.227	0.287	0.314	0.352	0.350	0.351	0.371	0.358	0.333	0.313	0.364	0.352	(40)	
(41)		taud	2.021	2.157	2.225	2.294	2.440	2.472	2.438	2.470	2.521	2.377	2.626	2.582	(41)	
(42)		Ebn at Noon	362	625	772	823	864	869	837	816	767	607	334	183	(42)	
(43)		Edh at Noon	28	62	87	103	97	96	97	86	67	51	26	16	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.14	0.71	2.21	3.80	5.12	5.66	5.17	3.95	2.26	0.84	0.20	0.06	(44)	
(45)		RadStd	0.01	0.14	0.23	0.34	0.52	0.49	0.48	0.38	0.23	0.16	0.03	0.01	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (44 neighbors)		+0.46	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page